

Assignment 3

CMPSC 487 Software Engineering and Design

Individual Assignment (50 points)

Assignment 3 Title: Pizza Customization Using Decorator Pattern

Objective: Design and implement a pizza customization system using the **Decorator Pattern**. The design should allow the dynamic addition of toppings and sauces to various types of pizzas. Students must design a class hierarchy for pizzas and sauces and demonstrate how the decorator pattern is used to add customization options.

Sample Decorator Pattern is attached to the Assignment.

Instructions:

1. Base Classes:

- Create a superclass Pizza with the following methods:
 - getDescription(): Returns a description of the pizza.
 - getCost(): Returns the base cost of the pizza.
- Implement at least three subclasses of Pizza:
 - MargheritaPizza
 - PepperoniPizza
 - VeggiePizza

Each subclass should define its own description and base cost.

2. Sauces:

- Create a subclass Sauce that also extends Pizza.
- Implement at least three types of sauces:
 - TomatoSauce
 - BarbecueSauce
 - PestoSauce

Each sauce should add a description and an extra cost when applied to a pizza.

3. Toppings Using Decorator Pattern:

- Implement toppings as decorators that dynamically add to the pizza. Examples of toppings:
 - CheeseTopping
 - OliveTopping
 - MushroomTopping
 - PepperTopping
- Each topping should add its own cost and update the description.

4. Submission Requirements:

- **Class Diagram:** Create and submit a UML class diagram showing the relationships between Pizza, Sauce, and the decorators.
- **Code Implementation:** Write Python code for the pizza customization system. Provide comments explaining how the Decorator Pattern is used.
- **Output Images:** Save some images of your output.

5. Example Output:

- **Take Input as your own style:** You can develop a Python console application and pressing different numbers will add different types of pizza.
 - **Order 1:** Margherita Pizza with Tomato Sauce, Cheese, and Mushrooms.
 - Description: "Margherita Pizza with Tomato Sauce, Cheese, and Mushrooms."
 - Total Cost: \$12.50
 - **Order 2:** Pepperoni Pizza with Barbecue Sauce and Olives.
 - Description: "Pepperoni Pizza with Barbecue Sauce and Olives."
 - Total Cost: \$14.00
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Evaluation Criteria (50 Points):

- **Class Diagram:** 20 points
 - **Code Implementation** (25 points)
 - **Base Classes and Subclasses:** 15 points
 - **Toppings Using Decorator Pattern:** 10 points
 - **Output similarity** (5 points)
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Late submissions will incur a penalty of 5% per day.